
Name:

Entry No.:

Use of unfair means will result in a F grade straight away. Also, unless stated otherwise, you may use results that we have proven in the class, but state them clearly whenever you use them.

1. [6 marks] We call an arbitrary set of formulas X *coherent* if there is no formula ϕ such that $X \vdash \phi$ and $X \vdash \neg\phi$. Prove that X is coherent *iff* every finite subset of X is coherent.

Mid-term exam (*Sept. 13, 25 marks, 2 hours*)

COL703/7203, Jul-Nov 2026

Name:

Entry No.:

Name:

Entry No.:

-
2. [5 marks] Let us say that a *positive resolution inference* is a resolution inference in which one of the hypothesis (i.e., one of the two clauses that are used in resolution) is a positive clause. We call a clause *positive* if it contains only positive literals. Now consider the following claim.

If a set of clauses S is unsatisfiable set of clauses, then S has a refutation containing only positive resolution inferences.

Is this claim true? You must clearly write “yes” or “no” first, and then give an argument to support your yes/no verdict. Answers with missing verdict or a missing/invalid/incomplete arguments will not get **any** marks.

Mid-term exam (*Sept. 13, 25 marks, 2 hours*)

COL703/7203, Jul-Nov 2026

Name:

Entry No.:

Name:

Entry No.:

-
3. [6 marks] Show that $\phi \vee \psi$ is consistent *iff* either ϕ is consistent or ψ is consistent. Use Hilbert's proof system for your derivations, but do not assume its soundness or completeness. Recall that ϕ is said to be *consistent* if $\not\vdash \neg\phi$. Here are the axioms of Hilbert's proof system.

- $\phi \rightarrow (\psi \rightarrow \phi)$ (A1)

- $(\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi))$ (A2)

- $(\neg\psi \rightarrow \neg\phi) \rightarrow ((\neg\psi \rightarrow \phi) \rightarrow \psi)$ (A3)

Mid-term exam (*Sept. 13, 25 marks, 2 hours*)

COL703/7203, Jul-Nov 2026

Name:

Entry No.:

Name:

Entry No.:

-
4. [5 marks] An adequate set of connectives and constants is a set such that for every formula in propositional logic, there is an equivalent formula with connectives and constants only from that set. Is $\{\top, \wedge, \vee, \rightarrow\}$ is an adequate set (where the symbol $\top$ denotes *true*)? You must clearly write “yes” or “no” first, and then give an argument to support your yes/no verdict. Answers with missing verdict or a missing/invalid/incomplete arguments will not get **any** marks.

Mid-term exam (*Sept. 13, 25 marks, 2 hours*)

COL703/7203, Jul-Nov 2026

Name:

Entry No.:

Name:

Entry No.:

5. [3 marks] Prove that *proof by contradiction* (PBC) is a derived rule (by using the basic rules of natural deduction to derive it). Recall that PBC says that if $\neg\phi$ leads to a contradiction, then ϕ can be deduced.

Mid-term exam (*Sept. 13, 25 marks, 2 hours*)

COL703/7203, Jul-Nov 2026

Name:

Entry No.:
